Supplementary Material: Gravity-Torque Momentum Management for a Three-Wheel Reaction-Wheel Inverted Pendulum

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S1. Plant parameters (nominal design point)

Parameter	Value	Unit
Cube edge L	0.150	m
Total mass M	1.41	kg
Pivot–CoM distance l	0.1465	m
Wheel axial inertia J_a	1.50e-04	kg m^2
Wheel speed limit Ω_{max}	1000	rad/s
Momentum budget h_{max}	0.150	N m s
Peak wheel torque τ_{max}	0.50	N m
Pivot friction b_p	1.0e-03	N m s/rad
Wheel friction b_w	1.0e-04	N m s/rad
Gyro noise (1σ)	2.0e-03	rad/s
Wheel Coulomb friction τ_c	1.5e-03	N m
Torque slew limit	50	N m/s
Controller rate f_s	500	Hz
Encoder resolution	4096	counts/rev
Gyro bias (1σ, per run)	5.0e-04	rad/s
Attitude noise (1σ)	5.0e-04	rad

The wheels are idealised discs whose rim reaches the seat distance ($r/d = 1$); a real wheel of finite thickness would have to be smaller. The Monte Carlo samples the fill ratio r/d with $\pm 20\%$ and caps it at 0.95, so most perturbed plants sit at the cap and have a smaller momentum budget than the nominal one.

S2. Tuned gains

Gains of the paper (Table I), tuned with the cost of Sec. IV-H, and of the survival-first re-tuning reported in Sec. VI-D. LQR/MPC: state weights q_θ, q_ω, q_h and input weight r; FBL+LQR: outer-loop weights; SMC: Λ, K, boundary layer ϕ; backstepping: k_1, k_2 ($\Gamma = I$, not tuned); PID: k_p, k_i, k_d. MPC: 25 steps of 40 ms, updates every 20 ms, OSQP. Momentum-management tilt clamp: 3°.

TABLE I
Gains used in the paper

Controller	Variant	Gains
LQR	classical	q_att= 42.39, q_rate= 0.00436, q_mom= 1.712, r= 1.008
MPC	classical	q_att= 7722, q_rate= 41.9, q_mom= 189, r= 1
FBL+LQR	classical	q_att= 4.249×10^6, q_rate= 60.9, r= 0.9978
SMC	classical	lam= 16.52, k_reach= 58.1, phi= 0.4218
Backstepping	classical	k1= 27.36, k2= 2.706
PID	classical	kp= 122.1, ki= 9.725, kd= 5.834
LQR	momentum aware	q_att= 42.39, q_rate= 0.00436, q_mom= 1.712, r= 1.008; k_h = 0.002081
FBL+LQR	momentum aware	q_att= 4.249×10^6, q_rate= 60.9, r= 0.9978; k_h = 0.1589
SMC	momentum aware	lam= 16.52, k_reach= 58.1, phi= 0.4218; k_h = 0.2293
Backstepping	momentum aware	k1= 27.36, k2= 2.706; k_h = 0.1775
PID	momentum aware	kp= 122.1, ki= 9.725, kd= 5.834; k_h = 0.2592
LQR	integral	q_att= 42.39, q_rate= 0.00436, q_mom= 1.712, r= 1.008; K_i = 0.7547
FBL+LQR	integral	q_att= 4.249×10^6, q_rate= 60.9, r= 0.9978; K_i = 0.3198
SMC	integral	lam= 16.52, k_reach= 58.1, phi= 0.4218; K_i = 0.1637
Backstepping	integral	k1= 27.36, k2= 2.706; K_i = 49.36
PID	integral	no separate arm: identical to the classical PID (K_i unused)

TABLE II
GAINS OF THE SURVIVAL-FIRST RE-TUNING

Controller	Variant	Gains
LQR	classical	q_att= 88.12, q_rate= 0.07368, q_mom= 1.624, r= 1.01
MPC	classical	q_att= 7722, q_rate= 41.9, q_mom= 189, r= 1
FBL+LQR	classical	q_att= 3.218×10^5, q_rate= 996.6, r= 0.9988
SMC	classical	lam= 18.91, k_reach= 11.88, phi= 0.337
Backstepping	classical	k1= 30.17, k2= 0.582
PID	classical	kp= 19.96, ki= 10.9, kd= 1.583
LQR	momentum aware	q_att= 88.12, q_rate= 0.07368, q_mom= 1.624, r= 1.01; $k_h = 0.02769$
FBL+LQR	momentum aware	q_att= 3.218×10^5, q_rate= 996.6, r= 0.9988; $k_h = 0.2563$
SMC	momentum aware	lam= 18.91, k_reach= 11.88, phi= 0.337; $k_h = 0.2585$
Backstepping	momentum aware	k1= 30.17, k2= 0.582; $k_h = 0.2869$
PID	momentum aware	kp= 19.96, ki= 10.9, kd= 1.583; $k_h = 0.2855$
LQR	integral	q_att= 88.12, q_rate= 0.07368, q_mom= 1.624, r= 1.01; $K_i = 0.6531$
FBL+LQR	integral	q_att= 3.218×10^5, q_rate= 996.6, r= 0.9988; $K_i = 0.0165$
SMC	integral	lam= 18.91, k_reach= 11.88, phi= 0.337; $K_i = 0.02561$
Backstepping	integral	k1= 30.17, k2= 0.582; $K_i = 0.02738$
PID	integral	no separate arm: identical to the classical PID (K_i unused)